

A general Łojasiewicz-Simon inequality

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Uniqueness problems in geometric analysis

Many interesting problems in geometric analysis have to deal with uniqueness and classification of possible singularities. To name a few:

- Uniqueness of tangent cone: on Einstein manifolds, (stationary) varifolds;
- Uniqueness/rigidity of tangent flows in MCF/RF: conical singularities, cylindrical singularities, solitons;
- Dynamical (in)stability of certain metrics (along geometric flows).

Uniqueness problems in geometric analysis

While each uniqueness problem has its own geometric context, many can be formulated as a search for the unique minimizer of a particular geometric functional.

And under suitable geometric modifications, the minimizers arise as critical points of gradient flows induced by the corresponding functionals. Here are some well-known examples:

- $\mathcal{F}$ -functional and self-shrinkers in MCF;
- λ -functional and Ricci shrinkers in RF;
- (The) tangent map of energy-minimizing maps on compact real-analytic manifolds;
- μ -functional and shrinker metric of normalized KRF on compact KE manifolds.

Classical Łojasiewicz theorem

We are particularly interested in uniqueness problems that can be reformulated as **gradient flows** with respect to certain functionals. And it turns out that the **Łojasiewicz theorem** provides a nice framework for such problems.

Theorem (Łojasiewicz)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a differentiable function, satisfying the following “**Łojasiewicz inequality**”: for any critical point p of f , there exists neighborhood U_p of p , constant $\beta \in (\frac{1}{2}, 1)$ such that for all $x \in U_p$,

$$|f(x) - f(p)|^\beta \leq |\nabla f(x)|. \quad (1)$$

Let $x : [0, \infty) \rightarrow \mathbb{R}^n$ be a curve with velocity $x' = -\nabla f$. Suppose $x(t)$ has a limit point x_∞ . Then x_∞ is the **unique limit point** of $x(\cdot)$ and also a **critical point** for f . In particular, when f is real-analytic, (1) is **automatically satisfied** for all $\beta \in [\frac{1}{2}, 1)$.

Classical Łojasiewicz theorem

Proof.

We prove that x_∞ is the unique limit point of $x(t)$ by showing that the curve x has finite length. Note that

$$\frac{d}{dt}f(x(t)) = \nabla f(x(t)) \cdot x'(t) = -|\nabla f(x(t))|^2 \quad (2)$$

and

$$f(x(t)) \leq Ct^{\frac{1}{1-2\beta}}. \quad (3)$$

Assume $f(x_\infty) = 0$. By (4.1) we have

$$\frac{d}{dt}f(x(t)) \leq -f(x(t))^{2\beta} \quad (4)$$

for $x(t)$ near x_∞ . □

Classical Łojasiewicz theorem

Proof.

Then

$$\int_1^\infty |\nabla f|(x(s)) ds = \int_1^\infty \sqrt{-\frac{d}{ds} f(x(s))} ds \quad (5)$$

$$\leq \left(- \int_1^\infty f' s^{1+\epsilon} ds \right)^{1/2} \left(\int_1^\infty s^{-1-\epsilon} ds \right)^{1/2}. \quad (6)$$

Integrating by parts gives

$$\int_1^T f' s^{1+\epsilon} ds = (fs^{1+\epsilon})|_1^T - (1+\epsilon) \int_1^T fs^\epsilon ds. \quad (7)$$

Choosing ϵ sufficiently small depending on β by (3), we see that (6) is bounded independent of T and hence the curve $x(t)$ has finite length. $\square$

Main scope of today's talk

- In this talk, we investigate how the classical Łojasiewicz theorem is applied in uniqueness problems in geometric analysis. In particular, we discuss the “**general Łojasiewicz-Simon inequality**” and relevant techniques introduced in Colding-Minicozzi's 2014 paper, *On uniqueness of tangent cones for Einstein manifolds*.
- Major challenge: validating Łojasiewicz theorem on infinite-dimensional function spaces (Banach spaces).
- If time permits, we will discuss the use of “slice theorem” in the Łojasiewicz-Simon argument.
- For convenience, we begin with preliminaries in Banach space.

Definition (Differentiability, C^p)

Let X, Y be two Banach spaces, $f : X \rightarrow Y$ a continuous mapping, $B(X, Y)$ denotes the space of bounded linear maps from X to Y .

$f : X \rightarrow Y$ is said to be (Frechet) **differentiable** at $x_0 \in X$ if there exists $A \in B(X, Y)$ such that

$$\|f(x_0 + u) - f(x_0) - Au\| = o(r) \quad (8)$$

for $\|u\| \leq r$ as $r \rightarrow 0$. Denote such A by $Df(x_0)$. If $Df(x_0)(\cdot)$ is also continuous as a map $x_0 \mapsto B(X, Y)$, then f is said to be of C^1 . Indeed we can inductively define C^p -maps.

Definition (Real-analytic maps between Banach spaces)

If X, Y are real Banach spaces, then f is said to be **real-analytic** in $U \subset X$ if there exists a family of maps a_k , $k = 0, 1, \dots$ such that each a_k is a symmetric, k -linear map from $U^{\otimes k} = U \otimes \cdots \otimes U$ to Y and the power series

$$f(x) = \sum_{k \geq 0} a_k(x, \dots, x)$$

converges absolutely for all $x \in U$.

Calculus on Banach space

Just as in Euclidean spaces, Banach spaces also admit implicit function theorem.

Theorem (Implicit function theorem: Banach space version)

Let X, Y, Z be (real or complex) Banach spaces, f a continuous mapping of an open set $U \subset X \times Y \rightarrow Z$. Assume that f has a Frechet derivative with respect to x , $Df(x)(\cdot, \cdot)$, which is continuous in U (i.e. varies continuously with respect to the basepoint). Let $(x_0, y_0) \in U$ and $f(x_0, y_0) = 0$. If $A = Df_x(x_0, y_0)$ is an isomorphism from X onto Z , then:

- ① There exists a ball $B_r(y_0)$ and a unique continuous map $u : B_r(y_0) \rightarrow X$ such that $u(y_0) = x_0$ and $f(u(y), y) \equiv 0$;
- ② If f is of class C^1 , then $u(y)$ is of class C^1 and

$$u_y(y) = -[Df_x(u(y), y)]^{-1} \cdot f_y(u(y), y); \quad (9)$$

- ③ $u(y) \in C^p$ if $f \in C^p$.

Proof.

Identical to the finite-dimensional case: use the contraction principle and integrated mean value property. □

Since the proof only makes use of the contraction principle, the implicit solution u is obtained via taking a limit to iterated compositions of function of class C^p . As the uniform limit of analytic maps is also analytic, we can conclude that the implicit function theorem identically holds for the holomorphic/real-analytic class of functions too, with their corresponding implicit solutions belonging to the same class of differentiability.

Standard elliptic theory

Definition (Fredholm)

A bounded linear operator between Banach spaces is **Fredholm** if it has finite-dimensional kernel and cokernel.

Let M be a smooth, **compact** manifold without boundary, E and F real smooth vector bundles over M .

Theorem (Schauder estimates)

Let $P : C^\infty(E) \rightarrow C^\infty(F)$ be a linear elliptic differential operator of order k . Given a constant $\alpha \in (0, 1)$ and an integer $l \geq 0$ there are constants c_1, c_2, c_3 such that for every $u \in C^{k+l,\alpha}(E)$

$$\|u\|_{C^{k+l,\alpha}} \leq c_1 \|Pu\|_{C^{l,\alpha}} + c_2 \|u\|_{C(E)} \leq c_3 \|u\|_{C^{k+l,\alpha}} \quad (10)$$

where $c_2 = 0$ for $u \perp \ker P$. In particular, $P : C^{k+l,\alpha}(E) \rightarrow C^{k,\alpha}(F)$ is an isomorphism between Banach spaces.

Now we state the existence theorem for elliptic operators. The key to the existence argument is to utilize the **Fredholm property** of elliptic operators.

Recall that in linear algebra, given a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^k$ the equation $Lx = y$ is solvable if and only if y is orthogonal to $\ker L^*$, which consequently yields the orthogonal decomposition

$$\mathbb{R}^k = L(\mathbb{R}^n) \oplus \ker L^*. \quad (11)$$

The following theorem, so called **Fredholm alternative**, is essentially the infinite-dimensional analogue of the image-cokernel decomposition as (11).

Standard elliptic theory

Theorem (Fredholm alternative)

Let $P : C^\infty(E) \rightarrow C^\infty(F)$ be a linear elliptic differential operator of order k . Then the following properties hold:

- Both $\ker P$ and $\ker P^*$ are finite-dimensional subspaces of C^∞ .
- If $f \in W^{l,2}(F)$, then the equation $Pu = f$ has a solution $u \in W^{k+l,2}$ if and only if f is orthogonal to $\ker P^*$; u is unique when it is required to be orthogonal (with respect to L^2) to $\ker P$.
- If $E = F$, then the eigenspace $\ker(P - \lambda I)$ is of finite dimension.

Consequently, we have the following orthogonal decompositions:

$$W^{l,p}(F) = P(W^{l+k,p}(E)) \oplus \ker P^* \quad (12)$$

$$C^{l,\alpha}(F) = P(C^{k+l,\alpha}(E)) \oplus \ker P^* \quad (13)$$

$$C^\infty(F) = P(C^\infty(E)) \oplus \ker P^*. \quad (14)$$

If P is **overdetermined elliptic**, i.e. its principal symbols are all injective, then (12)-(14) are valid when each $\ker P^*$ is replaced by the intersection of $\ker P^*$ with $W^{l,p}(F)$, $C^{l,\alpha}(F)$, $C^\infty(F)$, respectively.

Generalized Łojasiewicz-Simon inequality

- In [Sim83] L. Simon establishes a theoretical framework for the use of Łojasiewicz theorem in the context of geometric functionals defined on infinite-dimensional function spaces via **Lyapunov-Schmidt reduction**.
- While Simon's work is already a vast generalization of Łojasiewicz's original theorem, it requires uniform convexity conditions on the given functional, which may be absent in other settings.
- We introduce the **general Łojasiewicz-Simon inequality** suggested in [CM14], which is built upon relatively less restrictive requirements on the considered functional.
- Colding-Minicozzi's Łojasiewicz-Simon inequality framework is also employed in Deruelle-Ozuch's recent work [DO20], subject to the functional λ_{ALE} defined on a **noncompact** ALE manifold.

Statement of the general Łojasiewicz-Simon inequality

Suppose that we have:

- 1 A closed subspace E of L^2 maps to a finite-dimensional vector space (i.e. the target of the maps in E is a finite dimensional vector space) and a real-analytic functional G (with respect to Hölder norms) defined on a neighborhood $\mathcal{O}_E$ of $0 \in C^{2,\beta} \cap E$.
- 2 The gradient of G is a C^1 map (with respect to Hölder norms) $\nabla G : \mathcal{O}_E \rightarrow C^\beta \cap E$ with $\nabla G(0) = 0$ and Lipschitz in L^2 :

$$\|\nabla G(x) - \nabla G(y)\|_{L^2} \leq C \|x - y\|_{W^{2,2}}. \quad (15)$$

- 3 The linearization L of ∇G at 0 is symmetric, bounded from $C^{2,\beta} \cap E$ to $C^\beta \cap E$ and from $W^{2,2} \cap E$ to $L^2 \cap E$, and is **Fredholm** from $C^{2,\beta} \cap E$ to $C^\beta \cap E$.

Statement of the general Łojasiewicz-Simon inequality

Theorem (General Łojasiewicz-Simon inequality)

If G satisfies (1), (2) and (3), there exists $\alpha \in (0, 1)$ so that for all $x \in E$ sufficiently small,

$$|G(x) - G(0)|^{2-\alpha} \leq \|\nabla G(x)\|_{L^2}^2. \quad (16)$$

In [Sim83], C^1 -ness of ∇G and **Fredholm property of L** is guaranteed by the ellipticity of Euler-Lagrange operator for G . But here we don't make such an assumption on G , for there is another machinery (**slice theorem**) to obtain the Fredholm property indirectly.

Lyapunov-Schmidt reduction

To apply the classical Łojasiewicz theorem, we need to restrict our attention to some **finite-dimensional subspace** of the considered function space (why?). To do this, we need the following lemma.

Let Π_K denote the projection map onto the finite-dimensional (since L is Fredholm by assumption (3)) subspace $K := \ker L$.

Lemma (Lyapunov-Schmidt reduction**)

Define the mapping $\mathcal{N} = \nabla G + \Pi_K$. There is an open set $\mathcal{O} \subset C^\beta \cap E$ about 0 and a map $\Phi : \mathcal{O} \rightarrow C^{2,\beta} \cap E$ with $\Phi(0) = 0$ so that

- Φ is the inverse of $\mathcal{N}$;
- $\|\Phi(x)\|_{C^{2,\beta}} \leq C \|x\|_{C^\beta}$ and $\|\Phi(x) - \Phi(y)\|_{W^{2,2}} \leq C \|x - y\|_{L^2}$;
- The function $f = G \circ \Phi$ is analytic.

**The lemma above is called Lyapunov-Schmidt reduction because $f_K := f|_K$ is a real-analytic functional defined on a finite-dimensional vector space K fully capturing the critical points within, which makes it possible for us to apply the Łojasiewicz inequality on it.

Lyapunov-Schmidt reduction: proof

- By assumption, the mapping $\mathcal{N}$ is C^1 from $C^{2,\beta} \cap E$ to $C^\beta \cap E$ and its Frechet derivative at 0 is

$$d\mathcal{N}_0 = L + \Pi_K. \quad (17)$$

- Since L is Fredholm by assumption and Π_K is compact (it is of finite rank), the sum $L + \Pi_K$ is also Fredholm. In addition, since both L and Π_K are symmetric, so is their sum and, thus, $d\mathcal{N}_0$ is an isomorphism if and only if it is injective; and the injectivity follows from the definition of L and Π_K .
- Hence we conclude that $d\mathcal{N}_0$ is an isomorphism from $C^{2,\beta} \cap E$ to $C^\beta \cap E$ and the inverse $[d\mathcal{N}_0]^{-1}$ is a bounded linear mapping from $C^\beta \cap E$ to $C^{2,\beta} \cap E$ by the closed graph theorem.

Lyapunov-Schmidt reduction: proof

- Then the implicit function theorem gives an open set $\mathcal{O} \in C^\beta \cap E$ about 0 and a C^1 inverse map $\Phi : \mathcal{O} \rightarrow C^{2,\beta} \cap E$ with $\Phi(0) = 0$; the Frechet derivative of Φ is continuous and is given by

$$d\Phi_y = [d\mathcal{N}_{\Phi(y)}]^{-1}. \quad (18)$$

- Since Φ is of C^1 , the integral mean value theorem on Banach spaces gives a constant C so that for $x, y \in \mathcal{O}$

$$\|\Phi(x) - \Phi(y)\|_{C^{2,\beta}} \leq C \|x - y\|_{C^\beta}. \quad (19)$$

Using this with $y = \Phi(y) = 0$ gives $\|\Phi(x)\|_{C^{2,\beta}} \leq C \|x\|_{C^\beta}$. The Lipschitz bound for Φ as a map from L^2 to $W^{2,2}$ follows in the same way using the $W^{2,2}$ estimate for ∇G given in the assumption and the trivial boundedness of Π_K on L^2 .

Lyapunov-Schmidt reduction: proof

- Finally by implicit function theorem, Φ is also real-analytic since it is the inverse of real-analytic functional $\mathcal{N}$. Hence f is also real-analytic.

Important estimates

Lipschitz bound on Φ

$$\|\Phi(x) - \Phi(y)\|_{W^{2,2}} \leq C \|x - y\|_{L^2}$$

yields the following estimates, leading up to Łojasiewicz inequality for ∇G .

Lemma

There exists C such that for every sufficiently small $x \in C^{2,\beta} \cap E$

$$\|\nabla f(\Pi_K(x))\|_{L^2}^2 \leq C \|\nabla G(x)\|_{L^2}^2. \quad (20)$$

Lemma

There exists C so that for every sufficiently small $x \in C^{2,\beta} \cap E$

$$|G(x) - f(\Pi_K(x))| \leq C \|\nabla G(x)\|_{L^2}^2. \quad (21)$$

General Łojasiewicz-Simon inequality: proof

Let $x \in E$ be sufficiently small. Applying (20) and the finite dimensional Łojasiewicz inequality to f_K gives

$$\begin{aligned} C \|\nabla G(x)\|_{L^2}^2 &\geq \|\nabla f(\Pi_K(x))\|_{L^2}^2 \geq |\nabla f_K(\Pi_K(x))|^2 \\ &\geq |f_K(\Pi_K(x)) - f_K(0)|^{2-\alpha} \\ &= |f(\Pi_K(x)) - G(0)|^{2-\alpha}. \end{aligned}$$

Then the desired inequality follows from the triangle inequality and (21).

Slice theorem: Fredholm property of L

- In [CM14], the functional $\mathcal{R}$ under consideration acts on a space of pairs of metric and function $\mathcal{A}$, which is by nature an **infinite-dimensional space**.
- However, the general Łojasiewicz inequality requires the linearization L of $\nabla\mathcal{R}$ to be **Fredholm**, which does not apply to $\mathcal{R}$ directly, for there is the **diffeomorphism group** $\mathcal{D}$ acting on the kernel.
- To remedy this issue, we need to mod out the action of the diffeomorphism group using the **slice theorem**. By doing so, we have $\nabla\mathcal{R}$ **acting on a finite-dimensional slice of** $\ker L$ and therefore it is possible to apply the Łojasiewicz-Simon inequality.
- Here we explain how exactly the slice theorem fits into the general scheme of Łojasiewicz-Simon argument. For the notations used, please refer to [CM14].

Slice theorem on compact manifold

Theorem (Ebin-Palais slice theorem)

Let N be a closed Riemannian manifold, $\mathcal{R}(N)$ the space of Riemannian metrics with designated regularity $(W^{k,p}, C^{k,\alpha})$. Let $\mathcal{D}$ denote the diffeomorphism group on N . Given a metric $g \in \mathcal{R}(M)$, there exists a submanifold $S_g \subset \mathcal{R}(M)$ (of Banach manifold) containing g such that the followings hold:

- For any $\phi \in \text{Stab}(g)$, $\phi \cdot S_g \subset S_g$.
- If $\phi \in \mathcal{D}$ with $\phi \cdot S_g \cap S_g \neq \emptyset$, then $\phi \in \text{Stab}(g)$.
- There exists an open neighborhood U of the identity right-coset in $\mathcal{D}/\text{Stab}(g)$, and a cross section $\chi : U \rightarrow \mathcal{D}$, such that the map $F : U \times S_g \rightarrow \mathcal{R}(N)$ defined by

$$F(u, s) := \chi(u) \cdot s, \quad (22)$$

is a diffeomorphism onto an open neighborhood of g in $\mathcal{R}(M)$.

Slice theorem: idea

- Slice theorem tells us that locally, $\mathcal{R}(M) \cong \mathcal{D}(M) \times \text{'slice'}$.
- Now we want to show that, under certain ellipticity assumptions, the 'slice' – an open subset orthogonal to the action of diffeomorphism group – is in fact finite-dimensional.
- Since the Łojasiewicz-Simon inequality is available on the finite-dimensional slice (given the analyticity of considered functional), we can prove one on the entire neighborhood.

- In [CM14], the authors construct a functional $\mathcal{R} : \mathcal{A}_1 \rightarrow \mathbb{R}$ where it is set to satisfy the Łojasiewicz inequality

$$|\mathcal{R}(g, w) - \mathcal{R}(b_\infty^{-2}g_0, b_\infty)|^{2-\alpha} \leq |\nabla_1 \mathcal{R}|^2(g, w)$$

for $(g, w) \in \mathcal{A}_1$, where $\mathcal{A}_1$ is a given function space consisting of pairs of metric and function defined on a compact manifold N .

- Here $(b_\infty^{-2}g_0, b_\infty)$ is a critical point for $\mathcal{R}$ on $\mathcal{A}_1$.
- $\mathcal{A}_1$ is an infinite-dimensional space of functions. To apply Łojasiewicz, one has to find a finite-dimensional slice of a neighborhood of $(b_\infty^{-2}g_0, b_\infty) \in \mathcal{A}_1$ to work with.

Tangent space to the action of $\mathcal{D}$

- Note that we are doing analysis on a compact manifold N . Consider $\mathcal{D}$ the space of $C^{3,\beta}$ diffeomorphisms on N . The group $\mathcal{D}$ acts by pullback on the space

$$\mathcal{A} := \{(g, w) | g: \text{a } C^{2,\beta}\text{-metric on } N \text{ and } w: \text{a } C^{2,\beta}\text{-function on } N\}.$$

- The **tangent space $\mathcal{T}_{\mathcal{D}}$ to this action** is given by

$$\mathcal{T}_{\mathcal{D}} = \{(\mathcal{L}_V g_0, 0) | V \text{ is a } C^{3,\beta} \text{ vector field}\}. \quad (23)$$

- Elements of $\mathcal{T}_{\mathcal{D}}$ are “tangent” to the action of $\mathcal{D}$ in the sense that it describes the movement of g_0 in the direction of Φ_V , the diffeomorphism infinitesimally generated by V .

Decomposition of tangent space

- Indeed, identify the **tangent space** $\mathcal{T}_{(g_0, w_0)}\mathcal{A} = \mathcal{T}$ of $\mathcal{A}$ at (g_0, w_0) as the space of pairs of symmetric $C^{2,\beta}$ 2-forms and functions on N .
- Consider a given metric g_0 on N , the divergence operator $\delta = \delta_{g_0}$, and the adjoint operator δ^* to δ . By direct computation, one can check that for a 1-form ξ ,

$$\delta^*\xi = -\frac{1}{2}\mathcal{L}_{\xi^\#}g_0, \quad (24)$$

which means that $\mathcal{T}_{\mathcal{D}} = \text{Im } \delta^*$.

Decomposition of tangent space

- Viewing δ as an overdetermined elliptic differential operator of order 1, Fredholm property of elliptic operators yields the decomposition

$$\mathcal{T} = \mathcal{T}_{\mathcal{D}} \oplus \mathcal{T}_1 \quad (25)$$

where

$$\mathcal{T}_1 := \ker \delta \cap \mathcal{T} = \{(h, v) \in C^{2,\beta} : \delta h = 0\} \quad (26)$$

is a finite-dimensional subspace.

- We are interested in the subspace $\mathcal{T}_1^0 \subset \mathcal{T}_1$ of variations that are tangent to $\mathcal{A}_1$, i.e. that preserve the weighed volume constant

$$\mathcal{T}^0 := \left\{ (h, w) \mid \int_N \left(\frac{1}{2} \text{Tr}_{g_0}(h) + 2 \right) d\mu_{g_0} = 0 \right\}, \quad (27)$$

$$\mathcal{T}_1^0 := \mathcal{T}_1 \cap \mathcal{T}^0. \quad (28)$$

Slice theorem on metrics in $\mathcal{A}$

- Note that $(b_\infty^{-2}g_0, b_\infty)$ is a critical point of $\mathcal{R}$ in the space $\mathcal{A}_1 := \{(g, w) \in \mathcal{A} : \int_N w \, d\mu_g = \text{vol}(\partial B_1(0))\}$ by construction of $\mathcal{R}$. So we apply the slice theorem near the metric $b_\infty^{-2}g_0$.
- By slice theorem, we have
 - a neighborhood $\tilde{U}_1$ of 0 in $\mathcal{T}_1$;
 - a neighborhood $\tilde{U}$ of $b_\infty^{-2}g_0$ in the space of $C^{2,\beta}$ **tensors**;
 - a neighborhood $\tilde{U}_0$ of $b_\infty^{-2}g_0$ in the orbit $\mathcal{D} \cdot b_\infty^{-2}g_0$;
 - a map $\chi : \tilde{U}_0 \rightarrow \mathcal{D}$ to a neighborhood of the identity map Id_N with $\chi(b_\infty^{-2}g_0) = Id_N$

such that the mapping

$$F(u, h) := \chi(u) \cdot (b_\infty^{-2}g_0 + h) \quad (29)$$

is a diffeomorphism from $\tilde{U}_0 \times \tilde{U}_1$ to $\tilde{U}$.

Slice theorem on $\mathcal{A}_1$: a visualization

- To visualize this, we can imagine $\tilde{U}_0$ as an interval of a flow-line crossing the basepoint $b_\infty^{-2}g_0$ in the space of metrics, $\tilde{U}_1$ as the subspace of the tangent space at $b_\infty^{-2}g_0$ that is normal to the direction of $\tilde{U}_0$.
- Under a suitable exponential map, $\tilde{U}_1$ is mapped to a submanifold normal to $\tilde{U}_0$, and their product will be diffeomorphic to a neighborhood of $b_\infty^{-2}g_0$ in the space of metrics.
- Hence instead of considering the entire tangent space to the metric, it suffices to check the finite-dimensional vector space $\mathcal{T}_1$ which can generate the other metrics by the action of the diffeomorphism group.

Łojasiewicz theorem on $\mathcal{A}_1$

- Notice that we have applied the slice theorem only on metric components in $\mathcal{A}_1$.
- Applying the slice theorem on $\mathcal{A}$ therefore yields the **surjective** map $F : \tilde{U}_0 \times U_1 \rightarrow U$ where $U_1 \subset \mathcal{T}_1$ is an open neighborhood of $(0, 0)$ and $U \subset \mathcal{A}$ an open neighborhood of $(b_\infty^{-2}g_0, b_\infty)$.
- Now we restrict our attention to $\mathcal{A}_1 \subset \mathcal{A}$ and its tangent space $\mathcal{T}_1^0$.

Łojasiewicz theorem on $\mathcal{A}_1$

Constructing an exponential map explicitly as below, we obtain a local analytic diffeomorphism between $\mathcal{T}_1^0$ and $\mathcal{A}_1$.

Lemma (Exponential map)

The analytic map $\exp$ on $\mathcal{T}_1^0$ given by

$$\exp(h, w) = \left(b_{\infty}^{-2} g_0 + h, \frac{\text{vol}(\partial B_1(0))}{A_1(b_{\infty}^{-2} g_0 + h, b_{\infty} e^w)} b_{\infty} e^w \right) \quad (30)$$

is a diffeomorphism from a neighborhood of 0 to a neighborhood of $(b_{\infty}^{-2} g_0, b_{\infty}^{-2})$ in $\mathcal{A}_1$.

Łojasiewicz theorem on $\mathcal{A}_1$

- Then it follows that

Corollary (Slice theorem $\mathcal{A}_1$)

There is a neighborhood U'_1 of $(b_\infty^{-2}g_0, b_\infty)$ in $\mathcal{A}_1$ and a constant C , so that for each $y \in U'_1$, there is $y_0 \in \mathcal{T}_1^0$ and $\eta \in \mathcal{D}$ so that $y = \eta \cdot \exp(y_0)$ and $\|\eta\|_{C^{3,\beta}} \leq C$.

- Above corollary makes it possible for us to prove a Łojasiewicz-Simon inequality of $\mathcal{R}$ by that of $G = \mathcal{R} \circ \exp$'s when restricted on $\mathcal{A}_1$: it suffices to prove that G satisfies the assumptions (1),(2),(3). For more details, refer to [CM14].